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Testing Heavy Neutral Leptons with Multi-Messenger Observations of NGC 1068

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The active galactic nucleus NGC 1068, recently identified as a high-energy neutrino source by IceCube, exhibits a GeV γ-ray flux that exceeds predictions from opaque corona cascade models. This tension suggests the existence of non-attenuated energy transport mechanisms from the compact core. We investigate whether heavy neutral leptons (HNLs), coupled to the Standard Model through mixing with muon neutrinos, can provide such a channel, and we derive the resulting *exclusionary constraint* on this HNL scenario from the multi-messenger observations of NGC 1068. We calculate the production of HNLs in charged-pion and muon decays associated with hadronic interactions, explicitly tracking the mass-dependent polarization redistribution. We then derive the spectrum from the visible $N \to \nu_\mu + e^+ + e^-$ decay channel and evaluate its secondary electromagnetic emission. We find that the predicted photon flux is at least 7 to 8 orders of magnitude below the observational upper limits once experimental bounds on $|U_{\mu N}|^2$ are applied, which excludes this HNL channel as the origin of the γ-ray excess. The polarization-tracking framework developed in this work is applicable not only to astrophysical studies but may also provide useful theoretical input for decay simulations, detection-efficiency analyses, and mass reconstruction in future accelerator-based HNL searches.

Keywords: Heavy neutral leptons; Inverse seesaw mechanism; Multi-messenger astronomy; Active galactic nuclei; NGC 1068.

1. Introduction

Multi-messenger astronomy has turned astrophysical accelerators into laboratories for physics beyond the Standard Model (BSM). The IceCube collaboration has identified the active galactic nucleus (AGN) NGC 1068 as a point source of neutrinos at energies around 1 TeV,[1] and the KM3NeT/ARCA telescope has reported a high-energy event candidate possibly reaching 220 PeV.[2] These observations indicate that hadronic acceleration operates in at least some AGN cores, and they constrain the neutrino flux that any transport mechanism in such a source must reproduce.

The same source is anomalous in γ-rays. The GeV γ-ray flux of NGC 1068 measured by Fermi-LAT exceeds the predictions of opaque corona cascade models, in which the γ rays are generated in the optically thick core and attenuated before escape.[3,4] The excess therefore suggests a transport channel that carries energy out of the compact core without being absorbed, and that converts this energy into observable photons only beyond the opaque region. Neutral, weakly interacting particles are plausible candidates for such a channel: they escape the core without electromagnetic attenuation and decay or scatter in the surrounding environment, where their secondaries radiate. The question we address is whether a specific BSM particle can play this role.

HNLs are motivated here on two levels. Theoretically, the non-zero but extremely small active neutrino masses require physics beyond the Standard Model.[5,6] HNLs are among the simplest mediators of neutrino mass and can simultaneously provide a dark-matter candidate and a target for laboratory searches.[7] In the Inverse Seesaw (ISS) mechanism the smallness of the light neutrino mass is controlled by a small lepton-number-violating parameter rather than by a very large Majorana mass, so that the mixing with active neutrinos can remain large enough to be phenomenologically relevant. Phenomenologically, a MeV-scale HNL mixed with the muon neutrino can be produced in the principal neutrino production chain of NGC 1068 (pion and muon decays) and possesses a visible decay channel, making it a plausible candidate for the transport channel of this source. Terrestrial experiments have not yet observed a definite HNL signal, but they constrain the mixing with the muon neutrino to $|U_{\mu N}|^2 \lesssim 10^{-4}$ for $3\,\mathrm{MeV} < m_N < 19.5\,\mathrm{MeV}$;[8] the bound is slightly weaker but of the same order over the 10 MeV to 30 MeV range considered here. These considerations motivate HNLs as a representative candidate; other neutral candidates, such as axion-like particles, have been studied separately for NGC 1068,[9] and we do not attempt a comparative survey here.

For HNLs in the mass range considered here, finite-mass effects and polarization play important roles in their production and decay. A quantitative assessment of their contribution to the observed γ-ray emission therefore requires a polarization-resolved treatment. In this work, we investigate the contribution of HNLs to the γ-ray emission of NGC 1068 and assess whether this mechanism can account for the observed excess. We compute the HNL source spectrum from $\pi^\pm \to \mu^\pm + N$ and $\mu^\pm \to N + e^\pm + \bar{\nu}_e$, separating the two polarization states and boosting to the laboratory frame while tracking the polarization redistribution. We then derive the electron/positron spectrum from the visible decay $N \to \nu_\mu + e^+ + e^-$, propagate these leptons in a representative environment of NGC 1068, and compute the resulting electromagnetic emission with GAMERA.[10] Comparing with the observed γ-ray flux, we find that the predicted photon flux lies at least 7 to 8 orders of magnitude below the observational upper limits once the experimental bound on $|U_{\mu N}|^2$ is applied. The HNL channel is therefore excluded as the origin of the γ-ray excess.

The paper is organized as follows. Section 2 gives the HNL coupling, the ISS parameter range, the single-Dirac approximation, and the decay channels. Section 3

presents the polarization-resolved calculations of HNL production in pion and muon decays and of the visible three-body decay $N \to \nu_\mu + e^+ + e^-$, and collects them into the electron source function injected into the radiation calculation. Section 4 applies this source function to the environment of NGC 1068, compares with observations, and gives the suppression budget and its robustness. Section 5 summarizes our conclusions.

2. ISS Model: Parameters and Prerequisites

HNLs interact with the SM only through their mixing with the active neutrinos. Therefore any process that produces ν_μ also produces N, with a rate multiplied by $|U_{\mu N}|^2$, provided that it is kinematically allowed. Section 3 uses this to compute N production in charged-pion and muon decay chains.

In the Type-I seesaw:[11–16]

$$m_\nu = \frac{m_D^2}{M_R},\ m_N = M_R,\ |U|^2 = \left(\frac{m_D}{M_R}\right)^2 \tag{1}$$

For an observable M_R the mixing is usually too small to be probed. The Inverse Seesaw (ISS)[11–16] introduces another fermion field S, and the mass matrix reads

$$\left(\overline{\nu_L}\ \overline{\nu_R^c}\ \overline{S^c}\right) \begin{pmatrix} 0 & m_D & 0 \\ m_D & 0 & M \\ 0 & M & \mu \end{pmatrix} \begin{pmatrix} \nu_L^c \\ \nu_R \\ S \end{pmatrix} \tag{2}$$

where μ is a small lepton-number-violating mass term. For $\mu \ll M$ and $\mu \ll m_D$,

$$m_\nu = \mu\frac{m_D^2}{M^2},\ m_N = M,\ |U|^2 = \frac{m_\nu}{\mu} \tag{3}$$

The key point is that the smallness of the light neutrino mass is controlled by the small parameter μ rather than by a very large M, so that for comparable neutrino masses the mixing can be larger than in the Type-I case.

For the parameters used here, $|U|^2$ can reach the 10^{-4} level: taking $m_\nu \sim 0.05\,\text{eV}$ and $\mu \sim$ keV gives $|U|^2 = m_\nu/\mu \sim 10^{-4}$, while $\mu \ll m_D, M$ is satisfied. This is comparable to the experimental bound[8] and motivates the benchmark mixing adopted below.

For small μ the heavy states form a nearly degenerate pseudo-Dirac pair.[17,18] Over the range $m_N \simeq 10\,\text{MeV}$ to $30\,\text{MeV}$ and $\mu \ll m_D, M$ considered here, the mass splitting is much smaller than m_N, so that production and decay cannot resolve the two states; summing over helicities is therefore equivalent to a single Dirac fermion. The helicity-resolved calculation of Sec. 3, with $s = \pm 1$, rests on this approximation.

Because the HNLs considered here are lighter than the pion, their production and decay are governed by low-energy weak processes. They decay only into leptons: a three-neutrino channel (invisible) and a $\nu_\mu e^+ e^-$ channel (visible). Section 3.3 gives the explicit value $BR \simeq 0.11$, which enters the suppression budget of Sec. 4.3.

3. Polarization-Resolved HNL Production and Decay

Under AGN conditions, high-energy cosmic rays undergo $p-p$ and $p-\gamma$ interactions, which are thought to be the main sources of secondary particles;[19] these interactions produce copious pions, whose decays subsequently generate high-energy neutrinos and γ-rays.[20] We consider the two primary production channels for muon-flavored neutrinos, charged-pion decay and muon decay, and then the visible decay of the HNL; polarization is tracked through all three stages.

Throughout this section, E_N and E denote laboratory-frame energies of the HNL and of the electron, respectively, while x and y are the corresponding dimensionless variables in the parent rest frame, defined where they first appear. The helicity label $s = \pm 1$ denotes the HNL spin projection along its momentum, with $s = +1$ for a right-handed HNL; the same convention is used in all three stages below.

3.1. *Production in pion decay*

The decay $\pi^+ \to \mu^+ + N$ is the HNL analogue of the process $\pi^+ \to \mu^+ + \nu_\mu$ as shown in Fig.1..[21]

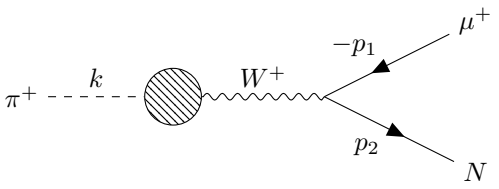


Fig. 1. Feynman diagram of π^+ decaying into μ^+ and massive HNLs.

By adapting the standard pion-decay calculation, we can find that the decay width to HNL is suppressed by the mixing angle and modified by a kinematic factor:

$$\frac{\Gamma_{\pi\to\mu N}}{\Gamma_{\pi\to\mu\nu}} = |U_{\mu N}|^2 \times \rho_\pi(m_N) \tag{4}$$

where

$$\rho_\pi(m_N) = \frac{(m_\pi^2(m_\mu^2 + m_N^2) - (m_\mu^2 - m_N^2)^2)\sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)}}{m_\mu^2(m_\pi^2 - m_\mu^2)^2} \tag{5}$$

$$\lambda(x, y, z) = x^2 + y^2 + z^2 - 2xy - 2yz - 2zx \tag{6}$$

The function ρ_π encodes the phase-space suppression.

The HNL polarization is important because it controls the angular dependence of the subsequent decay. In the pion rest frame we have

$$\Gamma_{(s)} \sim (m_\mu^2 + m_N^2)m_\pi^2 - (m_\mu^2 - m_N^2)^2 - s(m_\mu^2 - m_N^2)\sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)} \tag{7}$$

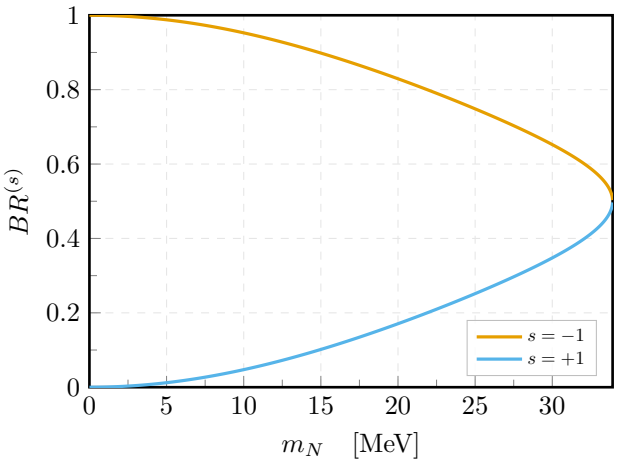


Fig. 2. Eq. (7) for different s. In pion decay, right-handed HNLs are more likely to be produced close to the phase space boundary.

where s is the polarization of HNL, and $s = +1$ for right-handed polarized HNLs. The m_N-dependent branching ratio for each polarization state in the pion rest frame is shown in Fig. 2.

Since the parent pion is a spin-zero scalar, the decay is isotropic in its rest frame. However, the polarization of the massive HNL needs to be carefully tracked, as the subsequent Lorentz boost to the laboratory frame induces a mass-dependent polarization redistribution. For a pion spectrum $J(E_\pi)$, we get

$$Q_N^{(s)}(E_N) = \int_{\lambda^-}^{\lambda^+} J(\frac{E_N}{x})|U_{\mu N}|^2 \rho_\pi \times r_{(s)}(x, E_N) \times \frac{dx}{(\lambda^+ - \lambda^-) \cdot x} \tag{8}$$

where

$$\lambda^\pm = \frac{m_\pi^2 - m_\mu^2 + m_N^2 \pm \sqrt{\lambda(m_\pi^2, m_\mu^2, m_N^2)}}{2m_\pi^2} \tag{9}$$

$$\cos\theta_r = \frac{x\,(m_\pi^2 - m_\mu^2 + m_N^2) - 2m_N^2}{x\sqrt{(1 - \frac{m_N^2}{E_N^2})\,\lambda(m_\pi^2, m_\mu^2, m_N^2)}} \tag{10}$$

$$r_{(s)}(x, E_N) = \frac{\Gamma_{(s)}\frac{1+\cos\theta_r}{2} + \Gamma_{(-s)}\frac{1-\cos\theta_r}{2}}{\Gamma_{(s)} + \Gamma_{(-s)}} \tag{11}$$

Here we assume $E_\pi \gg m_\pi$ to simplify the boost process. This resembles the SM neutrino case,[20] with modifications arising from HNL mass and helicity separation. The kinematic factor $\cos\theta_r$, illustrated in Fig. 3, controls the helicity redistribution during the Lorentz boost.

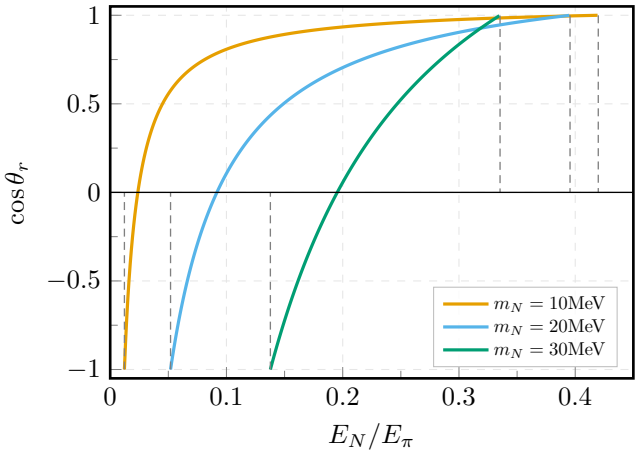


Fig. 3. Eq. (10) evaluated for different m_N values, which governs the helicity redistribution. $\cos\theta_r \approx 1$ holds across nearly the entire energy range when m_N is small, so the produced HNLs remain predominantly left-handed (cf. Fig. 2). As m_N approaches the kinematic threshold, the helicity flip is more significant and right-handed HNLs cannot be ignored.

3.2. *Production in muon decay*

Muons can decay into HNLs via a process analogous to their standard decay into ν_μ. The decay width of $\mu^- \to N e^- \bar{\nu}_e$ has the same form as that of the previously studied τ decay process[22]

$$\frac{\Gamma_{\mu\to N}}{\Gamma_{\mu\to\nu_\mu}} = |U_{\mu N}|^2 \times \rho_\mu(m_N) \tag{12}$$

where

$$\rho_\mu(m_N) = -r_N^8 + 8r_N^6 - 24r_N^4 \ln(r_N) - 8r_N^2 + 1, \quad r_N = m_N/m_\mu \tag{13}$$

The helicity-dependent spectrum of the HNL in the muon rest frame is given by

$$\frac{d\Gamma}{dE} = \frac{G_F^2}{24\pi^3}|U_{\mu N}|^2 \times |\mathbf{p}| \cdot [3(m_\mu^2 + m_N^2)E - 4m_\mu E^2 - 2m_N^2 m_\mu - s \cdot |\mathbf{p}|(m_N^2 + 3m_\mu^2 - 4m_\mu E)] \tag{14}$$

Several examples are plotted in Fig. 4.

A fully spin-correlated treatment could retain the polarization of the parent muons and the angular relation between their spin and decay products. Here we neglect parent-muon polarization as a simplifying approximation. This approximation isolates the HNL mass and polarization dependence but should not be interpreted as a demonstrated consequence of complete magnetic depolarization in the source.

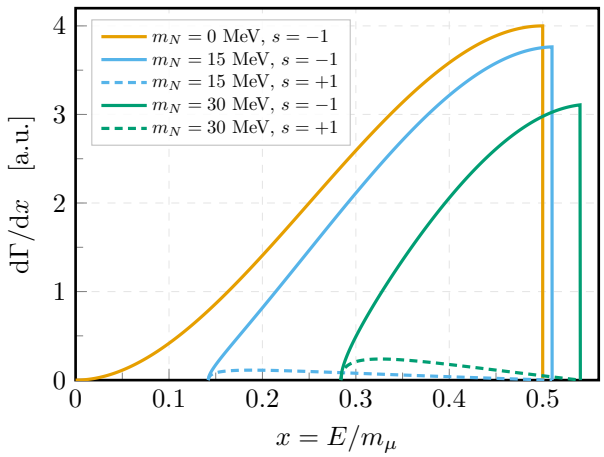


Fig. 4. Energy spectrum of HNL when muons decay into HNLs in the muon rest frame. The results are normalized to the $m_N = 0$ case, which corresponds to the SM neutrino and has no right-handed component. As m_N increases, the shape of the spectrum remains roughly the same, but the maximum energy increases while the total decay width decreases, yielding a increasing fraction of right-handed HNLs.

For a given muon spectrum, we have

$$Q_N^{(s)}(E_N) = \int_{\frac{m_N^2}{m_\mu^2}}^1 \frac{dy}{2y}\, Q_\mu(\frac{E_N}{y})|U_{\mu N}|^2 \rho_\mu \cdot \int_{x^-}^{x^+} \frac{dx}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \cdot \left(\frac{1+\cos\theta_r}{2} g_N^{(s)}(x) + \frac{1-\cos\theta_r}{2} g_N^{(-s)}(x)\right) \tag{15}$$

where

$$x^+ = \frac{m_N^2 + m_\mu^2}{2m_\mu^2},\ x^- = \frac{1}{2}\left(y + \frac{1}{y}\cdot\frac{m_N^2}{m_\mu^2}\right) \tag{16}$$

$$\cos\theta_0 = \frac{y - x}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \tag{17}$$

$$\cos\theta_r = \frac{1}{y\sqrt{1 - \frac{m_N^2}{E_N^2}}}\left(\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}} + x\cos\theta_0\right) \tag{18}$$

where $g_N^{(s)}(x)$ is normalized as

$$\int_{\frac{m_N}{m_\mu}}^{x^+} dx \sum_{s=\pm 1} g_N^{(s)}(x) = 1 \tag{19}$$

3.3. *Visible three-body decay*

Kinematically, low-mass HNLs in the ISS model can decay only into leptons. The possible final states are $\nu_\mu e^- e^+$ and $\nu_\mu \nu_l \bar{\nu}_l$. These processes have been well studied;[22–24] their decay widths are given by

$$\Gamma_{N\to 3\nu} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_{\mu N}|^2 \tag{20}$$

$$\Gamma_{N\to \nu e\bar{e}} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_{\mu N}|^2 \times C_1^e \tag{21}$$

where

$$C_1^e = \frac{1}{4}\left(1 - 4\sin^2\theta_W + 8\sin^4\theta_W\right) \tag{22}$$

To simplify the problem, we neglect the electron mass. The 3ν final state includes all neutrino flavor combinations.

An HNL produced in the central region can affect the electromagnetic signal only if its energy is ultimately transferred to visible charged particles, thus the branching ratio for an HNL to undergo a visible decay is

$$BR_{N\to \nu e\bar{e}} = \frac{C_1^e}{1 + C_1^e} \approx 0.11 \tag{23}$$

This implies a substantial energy loss, since only about one-tenth of the HNL decays produce charged leptons.

The angular and energy distributions of electrons/positrons are

$$\begin{aligned} g_e^{(1)}(x, \cos\theta) =& \frac{8}{C_1^e} \cdot x^2 \cdot \{[(4 - 16s_W^2 + 64s_W^4)x - (1 - 4s_W^2 + 28s_W^4)]\cos\theta \\ & - (4 - 16s_W^2 + 64s_W^4)x + (3 - 12s_W^2 + 36s_W^4)\} \end{aligned} \tag{24}$$

$$\begin{aligned} g_e^{(2)}(x, \cos\theta) =& \frac{16}{C_1^e} \cdot x^2 \cdot \{[(6 - 24s_W^2 + 32s_W^4)x - (3 - 12s_W^2 + 14s_W^4)]\cos\theta \\ & - (6 - 24s_W^2 + 32s_W^4)x + (3 - 12s_W^2 + 18s_W^4)\} \end{aligned} \tag{25}$$

Here, $g_e^{(1)}$ and $g_e^{(2)}$ represent the differential distributions for the outgoing electron (e^-) and positron (e^+), respectively, with $x = E_e/m_N$ and $s_W = \sin\theta_W$. The angle θ is defined between the outgoing lepton momentum and the parent HNL spin. To obtain the total injected lepton spectrum, we sum over both e^- and e^+:

$$\begin{aligned} Q_e^{(Lab)}(E) = BR_{N\to \nu e\bar{e}} \cdot & \int_E^{+\infty} dE_N\, Q_N^{(s)}(E_N) \\ & \cdot \int_{\frac{E}{2E_N}}^{\frac{1}{2}} \frac{dx}{xE_N} \sum_{i=1,2} g_e^{(i)}(x, s(\frac{E}{xE_N} - 1)) \end{aligned} \tag{26}$$

Evaluating the angular and energy integrals yields

$$Q_e^{(Lab)}(E) = BR_{N\to\nu e\bar{e}} \cdot \int_0^1 \frac{dy}{3y}\, Q_N^{(s)}(\frac{E}{y})[16y^3 - 27y^2 + 11 - s\cdot(32y^3 - 63y^2 + 36y - 5)] \tag{27}$$

where $y = E/E_N$. The dependence on the Weinberg angle θ_W cancels in the sum: in the massless limit $m_e \to 0$, helicity-flipping mass terms vanish, and the vector–axial-vector (V–A) interference terms are odd under charge conjugation ($e^- \leftrightarrow e^+$) and therefore cancel between the electron and positron yields, leaving the total spectral shape independent of $\sin^2\theta_W$.

4. HNL in NGC 1068

The observed neutrino flux of NGC 1068 is commonly modeled as originating from charged-pion decay.[4] Combining the analytical neutrino spectrum from pion decays[20] with theoretical models of NGC 1068,[4] we adopt the following parameterization for the pion spectrum:

$$\Phi_{\pi^\pm}(E) = \Phi_0 \cdot \left(\frac{E}{1\,\mathrm{GeV}}\right)^{-\Gamma} \cdot e^{-(E/E_c)^\beta} \tag{28}$$

with best-fit values

$$\Phi_0 = 4.7\times10^{-12}\,\mathrm{GeV}^{-1}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}, \Gamma = 0.1, E_c = 6.2\,\mathrm{GeV}, \beta = 0.3 \tag{29}$$

visualized in visualized in Fig. 5. The parameters are obtained by fitting this form to the theoretical neutrino flux of Ref.,[4] since the uncertainties and limited energy range of the IceCube data do not by themselves determine the pion spectrum. This spectrum provides the parent-particle input for the production calculation of Sec. 3.

4.1. *Decay length and the escape assumption*

For HNLs to mediate energy transport from the opaque core to a transparent external environment, their decay length must be sufficiently large. We have

$$l_{dec} = \frac{E_N}{m_N}\, c\, \frac{\hbar}{\Gamma_{tot}} = \frac{192\pi^3 c\hbar}{G_F^2(1+C_1^e)} \cdot \frac{E_N}{m_N^6 |U_{\mu N}|^2} \tag{30}$$

For example, setting $|U_{\mu N}|^2 = 10^{-5}$, $m_N = 20\,\mathrm{MeV}$, and $E_N = 1\,\mathrm{TeV}$ yields $l_{dec} \approx 0.4\,\mathrm{pc}$, which implies that HNLs are unlikely to escape into interstellar space, while some decays will still occur inside the optically opaque region.

Because $l_{dec} \propto E_N/(m_N^6|U_{\mu N}|^2)$, the decay length is extremely sensitive to m_N: over the 10 MeV to 30 MeV range it varies by more than two orders of magnitude. Since $l_{dec} \propto 1/|U_{\mu N}|^2$, at the experimental bound $|U_{\mu N}|^2 \lesssim 10^{-4}$ (the representative bound over the 10 MeV to 30 MeV range) the mixing is one order of magnitude larger than the value used above, so that l_{dec} is correspondingly shorter by about one order

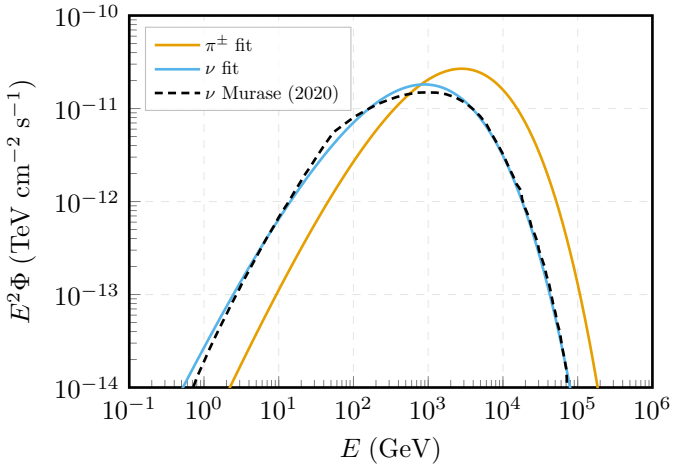


Fig. 5. Parametrization of the pion spectrum in NGC 1068 (Eq. (28)) for theoretical prediction.[4]

of magnitude (approximately 0.04 pc). Most HNLs then can decay inside the opaque core and their visible products are absorbed before escape. We therefore adopt the optimistic assumption that all HNLs decay beyond the core and that their energy escapes unattenuated; this assumption maximizes the predicted flux and makes the resulting exclusion conservative.

4.2. *Environment and radiation calculation*

We refer to the parameters of the AGN torus model.[25] For our approximation we set the magnetic field strength to 139 mG, dust temperature to 1500 K, ambient density to $2.30 \times 10^5\,\mathrm{cm}^{-3}$, and bolometric luminosity to $9.55 \times 10^{44}\,\mathrm{erg\,s}^{-1}$, and assume that electron-photon conversion happens at about 0.4 pc from the AGN core, corresponding to the inner edge of the torus in this model.

The electron/positron source function $Q_e^{(Lab)}(E)$ derived in Sec. 3.3 is injected into GAMERA,[10] which computes synchrotron, inverse-Compton (on the dust and CMB photon fields), and bremsstrahlung emission from these leptons in the adopted environment. Internal $\gamma\gamma$ absorption is neglected as an optimistic assumption; including it would further reduce the predicted flux. The resulting photon spectrum is shown in Fig. 7.

4.3. *Comparison with observations and the suppression budget*

Setting $|U_{\mu N}|^2 = 1$, the predicted photon flux lies three to four orders of magnitude below the observational upper limits. The residual suppression arises mainly from the small visible branching ratio $BR \simeq 0.11$, which contributes about one order of magnitude, and from the fraction of the parent pion or muon energy carried into the visible channel together with the radiation efficiency of the adopted environment,

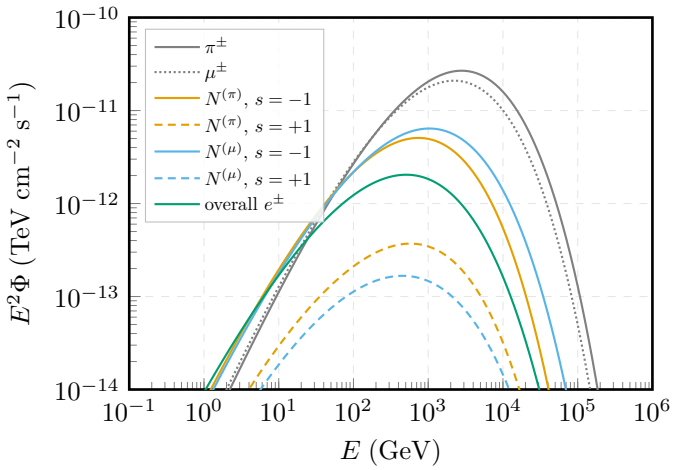


Fig. 6. HNL production source function from pion and muon decay for NGC 1068, using Eq. (28) as source pion spectrum. The polarization of muon is omitted. HNL mass is set to $m_N = 10\,\mathrm{MeV}$, and all HNL spectrums are scaled to $|U_{\mu N}|^2 = 1$ to better show the physics. For $m_N = 10\,\mathrm{MeV}$, right-handed neutrino flux is 1 to 2 orders of magnitude smaller than left-handed one.

which account for the remaining orders. The phase-space factors ρ_π, ρ_μ are of order unity over most of the 10 MeV to 30 MeV range and differ substantially from unity only near the kinematic endpoint, so they are not counted as a source of orders of magnitude. Applying the experimental bound $|U_{\mu N}|^2 \lesssim 10^{-4}$ (the representative value over this range) adds four more orders of magnitude, bringing the total suppression to seven to eight orders.

Two qualifications do not alter this result. First, the experimental bound on $|U_{\mu N}|^2$ is mass dependent and changing over the entire 10 MeV to 30 MeV range, although the difference is minor. Second, an accurate flux from a finite external region must account for the probability of decay within that region; the present treatment instead assumes that all HNLs deposit their visible products in the chosen zone, which makes the estimate optimistic rather than fully propagated. The first qualification affects the bound only at the same order, and the second can only reduce the predicted flux, so the exclusion is conservative. The comparison in Fig. 7 uses the Fermi-LAT point-source upper limits.[3] Within the range $m_N \simeq 10\,\mathrm{MeV}$ to 30 MeV, the channel is therefore excluded as an explanation of the γ-ray excess, although the laboratory limits on $|U_{\mu N}|^2$ are unchanged.

5. Conclusions

In this work, we have investigated whether heavy neutral leptons (HNLs) can provide an energy-transport mechanism to explain the GeV γ-ray excess observed in NGC 1068. We developed a polarization-resolved framework to calculate HNL production through charged-pion and muon decays, taking into account finite-mass

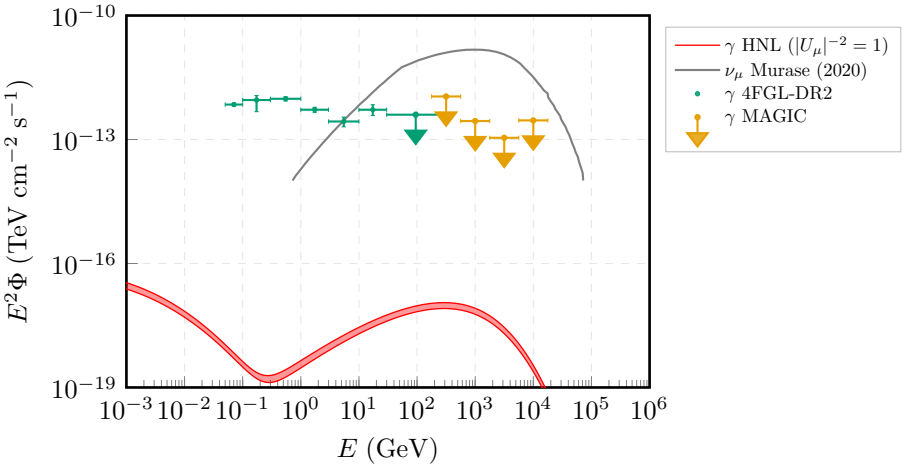


Fig. 7. Possible HNL generated photon spectrum compared with observations. m_N = 10 MeV to 30 MeV is shown as a range. Even ignoring the mixing angle, which is smaller than 10^{-4}, the flux is 3 to 4 orders of magnitude smaller than the observational upper limit, which means it is at least 7 to 8 orders of magnitude smaller and therefore can be completely ignored in observation.

effects and helicity redistribution under Lorentz boosts. The resulting HNL spectra were then used to derive the electron and positron distributions from the visible decay channel ($N \to \nu_\mu e^+ e^-$). By incorporating these distributions into a radiation model of NGC 1068, we calculated the associated γ-ray emission.

Our results show that, for HNL masses between 10 and 30 MeV, the predicted γ-ray flux is three to four orders of magnitude below the observational upper limits even when the mixing parameter is artificially set to $|U_{\mu N}|^2 = 1$. With existing laboratory constraints on the mixing parameter imposed, the predicted flux is suppressed by at least seven to eight orders of magnitude relative to these limits. Furthermore, our calculation adopts optimistic assumptions regarding HNL decay and photon escape, so a more realistic treatment of particle transport and attenuation would further reduce the predicted emission. These results indicate that the HNL scenario considered here cannot account for the observed GeV γ-ray excess in NGC 1068, although they do not impose new constraints on the HNL mixing parameter beyond existing laboratory bounds.

Although the predicted electromagnetic contribution is negligible, the polarization-resolved calculation provides a general framework for studying HNL production and decay in other astrophysical environments. The polarization-tracking method may also be relevant to accelerator-based HNL searches, such as those at the DUNE near detector and the proposed SHiP experiment,[26,27] as well as mass-reconstruction studies using liquid-argon time projection chambers (LArTPCs).[28] In particular, polarization-dependent decay distributions could provide useful theoretical input for decay simulations, detector-acceptance calculations,

and the reconstruction of HNL decay products.

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